Non-parametric Test

classmate

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The parametric test such as z-test, t-test, F-test are based on the assumption that the parent population is normal.

But when the population is not normally distributed and the sample data available may not be in interval and ratio scale of measurement. In such situations, the test of hypothesis is used in inferential statistics is non-parametric tests.

Definition:
Non-parametric tests are defined as those statistical tests of hypothesis in which no parameters are involved and which are based on order statistics.

Assumptions:
Some basic assumptions associated with non-parametric test are:

(i) A random sample is drawn from a population whose median is unknown.
(ii) The sample observation are independent.
(iii) The sample observation are measured at least in ordinal scale.
(iv) The sampled population is symmetric.
(v) The variable under study is continuous.
(vi) The lower order moments exists.

Advantages:

(i) Can be applied in qualitative as well as quantitative data.
(ii) It can be applied for small sample size.
(iii) It is less time consuming.
(iv) It needs no assumptions about the population from which sample is selected.
(v) It does not require complicated sampling theory.

Disadvantages:

(i) It cannot be used to estimate the parameters of population.
(ii) These test are less reliable and less powerful than parametric test.
(iii) These test are less efficient.

Uses of N.P. Test:

(i) For small sample size is given.
(ii) When assumptions of parametric procedures are not satisfied.
(iii) When quick data analysis is required.
(iv) When data show highly skewed.

Binomial Test:

It is a non-parametric test which is used to present data in either nominal or ordinal scale. It is used to test whether the binomial population has two distinct groups of two equal numbers outcomes or not.

let us consider a sample of size (n) is selected from Binomial population.

let $x_1, x_2, \ldots, x_n$ be independent sample of size (n) from binomial population having pmf $P(X=x) = {}^nC_x p^x q^{n-x}$ such that the probability associated with each trial is equal. x is the number of observations having certain characteristics and $n-x$ is number of observations having next characteristic then $p = x/n$.

A: Sample size $(n \leq 25)$

1. Formulation of hypothesis
   (a) $H_0: P = P_0$ (b) $H_1: P \neq P_0$ or $H_1: P_1 > P_0$ or $H_1: P_1 < P_0$
2. Test statistic: $x_0 = \min\{n_1, n_2\}$
3. Critical value: $p = P(X \leq x_0) = \sum_{x=0}^{x_0} {}^nC_x p^x q^{n-x}$

$$= \sum_{x=0}^{x_0} {}^nC_x (1/2)^n$$

4. Decision: If $p > \alpha$ % for OTT Accept H_0,
   $2p > \alpha$ for TTT Accept H_0
   Reject H_0 and Accept H_1 otherwise.

B: Large sample size $(n > 25)$:

2. Test statistic: $Z = \frac{x_0 - \mu}{\sigma} = \frac{x_0 - np}{\sqrt{npq}}$

   Since x_0 is discrete so that countinuity correction is made as,

$$x = \frac{(x_0 + 0.5) - np}{\sqrt{npq}}$$

If $x_0 < np$, use $+0.5$
If $x_0 > np$, use -0.5

3. Critical value: Z_α or $Z_{\alpha/2}$
4. Decision
5. Conclusion

Median Test:

It is used to test the significance difference between two independent distribution.

Let $x_1, x_2, \ldots, x_{n_1}$ and $y_1, y_2, \ldots, y_{n_2}$ be two independent samples of size n_1 and n_2 respectively be selected from continuous population with unknown medians Md_1 and Md_2 respectively.

① Formulation of hypothesis:

ⓐ H_0: $Md_1 = Md_2$

ⓑ H_1: $Md_1 \neq Md_2$ or $H_1: Md_1 > Md_2$ or $H_1: Md_1 < Md_2$

Ⓐ Small sample size ($n_1 \le 10$, $n_2 \le 10$)

Combine n_1 and n_2 such that $n = n_1 + n_2$ and obtain median of n observations. Find the number of observations in $x_i \le Md$ and denoted by 'a'.

② Test Statistic: $$P(A=a) = \frac{\binom{n_1}{a}\binom{n_2}{k-a}}{\binom{n}{k}}$$

where $a = 0, 1, 2, \ldots \min\{n_1, k\}$, $k = \frac{n}{2}$

③ Critical Value: Critical value is given by $P_0 = P(A \ge a)$

④ Decision: Accept H_0 if $P_0 > \alpha$ for OTT

if $2P_0 > \alpha$ for TTT

Ⓑ For large sample size ($n_1 > 10$, $n_2 > 10$)

Combine n_1 and n_2 such that $n = n_1 + n_2$ and obtain median of n observations. Find number of observations in $x_i \le Md$ and denoted by a.

Find number of observations in $y_i \le Md$ denoted by b. Also find the number of observations in $x_i > Md$ and y_i and, denoted by d respectively.

	No. of obs. $\le Md$	No. of obs. $> Md$	Total
Sample x	a	c	a+c
Sample y	b	d	b+d
Total	a+b	c+d	N

Test statistic: $$\chi^2 = \frac{N(ad-bc)^2}{(a+c)(b+d)(a+b)(c+d)} \sim \chi^2_{(1)}$$

If any cell frequency is less than 5.

$$\chi^2 = \frac{N\left[|ad-bc| - N/2\right]^2}{(a+c)(b+d)(a+b)(c+d)} \sim \chi^2_{(1)}$$

Example: Two independent samples are given below.

Sample I:	10	11	8	8	14	–	–
Sample II:	9	12	13	9	15	9	17

Test whether the two samples have come from the same population with respect to their medians. Use median test at 5% level of significance.

Soln:

Arranging the data, $n = 12$

8, 8, 9, 9, 9, 10, 11, 12, 13, 14, 15, 17

$Md = $ value of $\left(\frac{n+1}{2}\right)^{th}$ item $= 6.5^{th}$ item $= 10.5$

Now,

The number of observations in the first sample less than or equal to median (a) = 3

First sample size $(n_1) = 5$

Second sample size $(n_2) = 7$

$$k = \frac{n_1 + n_2}{2} = 6$$

Let Md_1 and Md_2 be median of I popn and II popn respectively.

1. Formulation of hypothesis:

(a) $H_0: Md_1 = Md_2$

(b) $H_1: Md_1 \neq Md_2$ (TTT)

2. Test statistic:

$$p(A = a) = \frac{({}^{n_1}C_a)({}^{n_2}C_{k-a})}{({}^{n}C_k)}, \quad a = 0, 1, 2, 3, 4, 5, (k)$$

3 Critical value:

$$P_0 = (A \geq a) = p(A \geq 3) = \sum \frac{({}^{5}C_a)({}^{7}C_{6-a})}{{}^{12}C_6}$$

$$P_0 = \frac{C(5,3)\ C(7,6-3)}{C(12,6)} + \frac{C(5,4)\,C(7,6-4)}{C(12,6)} + \frac{C(5,5)\,C(7,6-5)}{C(12,6)}$$

$P_0 = 0.5$

4. Decision:

Here, $2P_0 > \alpha$ i.e. $2 \times 0.5 > 0.05$

Accept H_0, Reject H_1.

5. Conclusion:

Two samples have come from same population with respect to median.

Example: An IQ test was given to a random sample of 15 male and 20 females students of a university. Their scores were recorded as follows:

Male: 56, 66, 62, 81, 75, 73, 83, 68, 48, 70, 60, 77, 86, 44, 72

Female: 63, 77, 65, 71, 74, 60, 76, 61, 67, 72, 64, 65, 55, 89, 45, 53, 68, 73, 50, 81

Solution:

Here, No. of male (n_1) = 15

No. of female (n_2) = 20

a(No. of obs. of male ≤ Md) = 7

b(No. of obs. of female ≤ Md) = 12

c(No. of obs of male > Md) = 8

d(No. of obs. of female > Md) = 8

	Sample I	Sample II	Total
No. of obs. ≤ Md	7(a)	12(b)	19 = a+b
No. of obs. > Md	8(c)	8(d)	16 = c+d
	a+c = 15	b+d = 20	35

Let Md_1 and Md_2 be median IQ of male and female respectively.

1. Formulation of hypothesis:

(a) $H_0: Md_1 = Md_2$

(b) $H_1: Md_1 \neq Md_2$

2. Test Statistic:

$$\chi^2 = \frac{N(ad-bc)^2}{(a+c)(b+d)(a+b)(c+d)} = \frac{35(7\times8-12\times8)^2}{15\times20\times19\times16}$$

$$\chi^2 = 0.16$$

3. Critical value: $\chi^2_{(1)} = 3.84$

4. Decision:

$\chi^2_{cal} < \chi^2_{tab}$, Accept H_0, Reject H_1 at $\alpha = 5\%$.

5. Conclusion:

IQ of male and female are same in the university.

Binomial Test:

Binomial test is used to test whether the binomial population has two distinct two groups of two equal numbers of outcomes or not.

Let $x_1, x_2, \ldots, x_n$ be independent sample of size n selected from Binomial population having pmf $p(X=x) = {}^nC_x p^x q^{n-x}$ such that probability associated with each trial is equal, x is number of observations having certain characteristic and $n-x$ is the number of observations having next characteristic then $p = x/n$.

1. Formulation of hypothesis:

(a) $H_0: P = P_0$

(b) $H_1: P \neq P_0$ (TTT) or $H_1: P > P_0$ (RTT) or $H_1: P < P_0$ (LTT)

(A) Small sample size ($n \le 25$)

2. Test statistic: $x_0 = \min\{n_1, n_2\}$

3. Critical value: $P = P(X \le W) = \sum_{x=0}^{x_0} {}^nC_x p^x q^{n-x}$

$$P = \sum_{x=0}^{x_0} {}^nC_x \left(\frac{1}{2}\right)^n$$

4. Decision: Accept H_0 at $\alpha\%$ level of significance if $P > \alpha\%$ for one tailed test and $2P > \alpha\%$ for TTT, otherwise reject H_0.

(B) For large sample ($n > 25$)

For large sample size, $x_0 \sim N(np, npq)$ i.e. x_0 is normally distributed with mean and variance npq.

Test statistic:

$$Z = \frac{x_0 - np}{\sqrt{npq}}$$

Since x_0 is discrete so that continuity correction is made as,

$$Z = \frac{(x_0 \pm 0.5) - np}{\sqrt{npq}}$$

use 0.5 if $x_0 < np$
use -0.5 if $x_0 > np$

Critical value:

obtain from table

$z_{\alpha/2}$ for TTT

z_{α} for OTT

Decision: Reject H_0 if $|Z|_{cal} > |Z|_{tab}$

Example: The following are defective (D) and non-defective (N) electronic items produced in the given order by a certain machine.

NNDDNDNNNDDNDDNNDNNDNNN

Test whether the defective and non-defective items are equally produced or not. Use binomial test at the 1% level of significance.

Solution:

Here,

Number of N(n_1) = 13

Number of D (n_2) = 9 (<25)

1. Formulation of hypothesis

(a) H_0: $p = 1/2$

(b) H_1: $p \neq 1/2$ (TTT)

2. Test Statistic:

$x_0 = \min\{n_1, n_2\} = \min\{13, 9\} = 9$

3. Critical Value:

$$p = \sum_{x=0}^{x_0} {}^{n}C_x (1/2)^n = \sum_{x=0}^{9} {}^{n}C_x (1/2)^n$$

$p = 0.262$

$\Rightarrow P_0 = 2p = 2 \times 0.262 = 0.524 = 52.4\%$ for TTT

4. Decision:

Here, $P_0 > \alpha\%$ where $\alpha = 1\%$

So, Accept H_0, Reject H_1

5. Conclusion:

Defective and non-defective items are ~~pe~~ equally produced.

Example: Out of 50 students willing to express opinion of laptop 30 expressed preferences to brand Dell and 20 for brand Lenovo. Use Binomial test to test the hypothesis that both brand of laptop are equally popular.

Solution:

Here,

No. of students prefer Dell laptop $(n_1) = 30$

No. of students preferring lenovo laptop $(n_2) = 20$

1. Formulation of hypothesis:

(a) $H_0: P = 1/2$

(b) $H_1: P \neq 1/2$ (TTT)

2. Test statistic:

$x_0 = \min\{n_1, n_2\} = \min\{20, 30\}$

$x_0 = 20$

$np = 50 \times 1/2 = 25 > x_0 \quad \therefore +0.5$

$$Z = \frac{(x_0 + 0.5) - np}{\sqrt{npq}} = -1.27$$

$|Z|_{cal} = |-1.27| = 1.27$

3. Critical Value:

$|Z_{\alpha/2}|_{tab} = 1.96$

4. Decision:

$1.27 = |Z|_{cal} < |Z|_{tab} = 1.96$

Reject H_1 and Accept H_0.

5. Conclusion:

Both brand of laptop are equally popular.

Sign Test:

A sign test is a very simple and easiest non-parametric test which applies to the median.

(A) One Sample Sign Test:

Case I: Sample size $n \leq 25$

Let a random sample $x_1, x_2, \ldots, x_n$ of small size $n \leq 25$ to be drawn from a population with pdf $f(x)$ with an unknown Median Md. Also, $P(x = Md) = 0$.

1. Formulation of hypothesis:

(a) $H_0: Md = Md_0$ i.e., the population median is Md_0 where Md_0 is some specified value of the population median.

(b) $H_1: Md \neq Md_0$ or $H_1: Md > M_0$ or $H_1: Md < Md_0$

2. Test Statistic:

Under H_0,

$$k = \min\{n(+), n(-)\}$$

3. Critical region:

Obtain P-value (P_0) from Binomial table

$$P_0 = \sum_{y=0}^{k} {}^{n}C_y \, p^y q^{n-y} \quad \text{where } q = 1-p$$

$$P_0 = (1/2)^n \sum_{y=0}^{n} {}^{n}C_y \quad \text{under } H_0$$

4. Decision:

(i) If $P_0 > \alpha$, Accept H_0 (For OTT)

(ii) If $2P_0 > \alpha$, Accept H_0 (For TTT)

Case II: Sample size $n > 25$

Test Statistic:
$$Z = \frac{k - np}{\sqrt{npq}}$$

$$\text{Corrected } z = \frac{(k \mp 0.5) - np}{\sqrt{npq}}$$

Critical value:

For pre-assigned level of significance α, we obtained the p-value (P_0).

Decision:

(i) If $P_0 > \alpha$, Accept H_0, for OTT

(ii) If $2P_0 > \alpha$, Accept H_0 for TTT

Example: The data of percentage of broken groundnut pods recorded from an experiment are given below: 9·8, 10·4, 11·5, 10·4, 8·5, 8·0, 10·5, 7·5, 8·8, 9·2. Test the hypothesis that the percentage of broken pods has the median 8. Use sign test.

Solution:

1. Formulation of hypothesis:

(a) H_0: $Md = 8$ i.e., the percentage of broken pods has the median 8.

(b) H_1: $Md \neq 8$ (TTT)

2. Test Statistic: Assign plus(+) sign, minus(−) sign and zero (0) to the given observations. i.e., find difference $x_i - 8$ and write their signs as follows:

1·8	2·4	3·5	2·4	0·5	0	2·5	−0·5	0·8	1·2
+	+	+	+	+		+	−	+	+

Here, $n(+) = 8$, $n(-) = 1$

Number of ties $(t) = 1$

Effective sample size $(n_e) = n - t = 10 - 1 = 9$

Therefore, the test statistic under H_0 is,

$k = \min\{n(+), n(-)\} = \min\{8, 1\} = 1$

3. Critical Value:

The p-value (P_0) is obtained as,

$$P_0 = \underset{P(Y \le k)}{P(Y \le 1)} = \left(\frac{1}{2}\right)^9 \sum_{y=0}^{1} {}^9C_y = 0.020$$

For two tailed test (TTT), $2P_0 = 2 \times 0.020 = 0.04$

$2P_0 = 4\%$

4. Decision:

Since $2P_0 = 4\%$ is less than $\alpha = 5\%$.

Therefore, we accept H_1 and reject H_0.

5. Conclusion:

Hence, we conclude that the percentage of broken pods has not median 8.

Example: A survey of the time records of 120 randomly selected employees was conducted and a plus (+) sign was recorded. If the employee was absent 15 days or more and, a minus (-) sign if the employee was absent less than 15 days. Total showed 80 minus and 40 plus signs. Use the sign test to test the hypothesis that the median number of days, Md, that an assembly line employees was absent for at least 15 days.

Solution:

Given,

Sample size $(n) = 120$ $\quad n > 25$, large sample case

Total number of plus signs $= n(+) = 40$

Total number of minus signs $= n(-) = 80$

1. Formulation of hypothesis

(a) Null hypothesis is $H_0: Md \geq 15$ i.e, the number of plus sign is at least as many as the number of minus signs.

(b) Alternative hypothesis $H_1: Md < 15$ (LTT) i.e., the number of plus sign is fewer than the number of minus sign.

2. Test statistic: Under H_0,

$$Z = \frac{(k \pm 0.5) - n/2}{1/2\sqrt{n}} \sim N(0,1)$$

$$= \frac{(k \pm 0.5) - np}{\sqrt{npq}}$$

where k = total number of less frequent sign $= n(+) = 40$

$$\therefore Z = \frac{(40 + 0.5) - 120 \times \frac{1}{2}}{\sqrt{120 \times \frac{1}{2} \times \frac{1}{2}}} = -3.562$$

3. Critical value: $P_0 = P(Z \leq -3.562) = 0.0002$

4. Decision: since $P_0 = 0.0002 < 0.05 = \alpha$

Accept H_1, reject H_0.

5. Conclusion: we conclude that there are significantly more minus sign than plus signs i.e., the median that an assembly line employees was absent is less than the claimed 15 days.

Run Test:

Case I: Small sample i.e., $n \le 40$

Let $x_1, x_2, \dots, x_n$ of size $n \le 40$ be drawn from non-normal population with pdf $f(x)$. Here, the basic assumption: We assume that the sample observations are measured in nominal or ordinal scale. Run test is used to test the randomness of the sample observations.

1. Formulation of hypothesis:

(a) H_0: The sample observations are in random order.

(b) H_1: The sample observations are not in random order (TTT).

OR, H_1: There is too few runs (LTT)

OR, H_1: There is too much runs (RTT)

2. Test statistic:

Under H_0,

R = Number of runs

where,

(i) Arrange the sample observations in ascending order of their magnitude.

(ii) Compute median (Md) from the arranged data.

(iii) Assign the observations x_i as follows

Assign A to the observations x_i, if $x_i > Md$

Assign B to the observations x_i, if $x_i < Md$

Assign O to the observations x_i, if $x_i = Md$

(iv) Obtain a sequence of A_s and B_s as given

AABBABBAAAB

(ii) Count the number of runs 'R' in the sequence of A and B.

3. Critical value:
For pre-assigned level of significance $\alpha = 5\%$, the critical value $\underline{r}$ and $\bar{r}$ can be obtained from the table of run test for different n_1 and n_2.

4. Decision:
(i) If $R \in (\underline{r}, \bar{r})$ Accept H_0
(ii) If $R \notin (\underline{r}, \bar{r})$ Accept H_1

Case II: Large Sample Case i.e, $n > 40$.
In this case, under H_0, the sampling distribution of the run 'R' is approximately normal distribution with mean $E(R) = \frac{n+2}{2}$ and $V(R) = \frac{n}{4}\left(\frac{n-2}{n-1}\right)$.

Therefore, the test statistic is z-statistic defined by,

$$Z = \frac{R - E(R)}{SD(R)} = \frac{R - \frac{n+2}{2}}{\sqrt{\frac{n}{4}\left(\frac{n-2}{n-1}\right)}} \sim N(0,1)$$

Critical Value: Obtain critical value or p-value (P_0), the probability associated with the value as extreme as the calculated value of z from the standard normal table.

Decision: (OTT)
(i) If $P_0 \leq \alpha$, Accept H_1, Reject H_0
If $P_0 > \alpha$, Accept H_0, Reject H_1

Mann-Whitney U-Test:

Mann-Whitney U-Test is a most powerful non-parametric test for testing the hypothesis of difference between two mean of two independent random samples drawn from same population or two different populations. This is a non-parametric alternative to the t-test and it is a very popular non-parametric test amongst the rank sum test for two independent random samples. The test procedure for of the Mann-Whitney U-test is given below:

Case I: When $n_1 \leq 20$ and $n_2 \leq 20$

Let two independent random samples $x_1, x_2, x_3, \ldots, x_{n_1}$ and $y_1, y_2, \ldots, y_{n_2}$ of respective size n_1 and n_2 both less than 20, be drawn from two population $F_X(x)$ and $F_Y(x)$ with respective prob. density functions $f_X(x)$ and $f_Y(x)$ with unknown $E(X)$ and $E(Y)$.

1. Formulation of Hypothesis

(a) H_0: $F_X(x) = F_Y(x)$ $\Rightarrow Md_1 = Md_2$ for all x i.e., the two random sample have been drawn from same population.

(b) H_1: $F_X(x) \neq F_Y(x)$ (TTT) $\Rightarrow Md_1 \neq Md_2$

or, H_1: $F_X(x) > F_Y(x)$ (RTT)

or, H_1: $F_X(x) < F_Y(x)$ (LTT)

2. Test Statistic:

Under H_0,

(i) combine the same observations of both sample so that $n = n_1 + n_2$.

(ii) Rank these n observations either from smallest to largest or from largest to smallest value. If the value of two or more observations are equal, then assign each of the tied observations by the average of their rank.

(iii) Determine sum of the ranks assigned to the values of the first sample and second sample separately and denote them by R_1 and R_2 respectively.

(iv) Compute U values defined in terms of R_1 and R_2 for each sample as,

$$U_1 = n_1 n_2 + \frac{n_1(n_1+1)}{2} - \sum R_1$$

$$U_2 = n_1 n_2 + \frac{n_2(n_2+1)}{2} - \sum R_2$$

so that $U_1 + U_2 = n_1 n_2$

(v) The test statistic is,

$$U_0 = \min\{U_1, U_2\}$$

3. Critical value:

For a pre-assigned level of significance α and the sample sizes n_1 and n_2. Obtain the critical value $U_{\alpha}(n_1, n_2)$ from the table of critical values of U in Mann-Whitney U test.

4. Decision:

(i) If $U_0 \le U_{\alpha/2}(n_1, n_2)$ ~~and~~ Reject H_0, Accept H_1

and, $U_0 \ge U_{1-\alpha/2}(n_1, n_2)$

(ii) If $U_{\alpha/2}, (n_1, n_2) < U_0 < U_{1-\alpha/2}(n_1, n_2)$ Accept H_0

(for TTT)

For left tailed test:

If $U_0 \leq U_\alpha (n_1, n_2)$ we reject H_0

If $U_0 > U_\alpha, (n_1, n_2)$ we accept H_0

For Right tailed test:

If $U_0 \geq U_{1-\alpha}, (n_1, n_2)$ reject H_0

If $U_0 \leq U_{1-\alpha} (n_1, n_2)$ Accept H_0

Alternatively,

Critical region:

(i) when $n_1 \leq 10$ and $n_2 \leq 10$ or when $n_1 \leq n_2$ and $3 \leq n_2 \leq 10$, we obtained the probability P_0 associated with the value as small as the observed U_0 for a pre-assigned level α and (n_1 and n_2) from the table of Mann-Whitney U test. That is, $P_0 = P(U \leq U_0)$

(ii) When $n_1 \leq n_2$ and $11 \leq n_2 \leq 20$, we obtain the critical value $U_\alpha(n_1, n_2)$ or $U_{\alpha/2}(n_1, n_2)$ from the table of critical values of U at a given level α and (n_1, n_2).

Decision:

(i) For OTT, if $P_0 \leq \alpha$, reject H_0, accept H_1

(ii) For TTT, if $2P_0 \leq \alpha$, reject H_0, accept H_1

Case II:

Large sample case i.e. when n_1 or $n_2 > 20$ or when $n_1 \leq n_2 > 20$.

Test statistic:

The sampling distribution of U_0 is approximately normal with mean,

$$E(U) = \mu_U = \frac{n_1 n_2}{2}$$

$$\sigma_U^2 = \frac{n_1 n_2 (n_1 + n_2 + 1)}{12}$$

Therefore, the test statistic is,

$$Z = \frac{U_0 - \mu_U}{\sigma_U} = \frac{U_0 - \frac{n_1 n_2}{2}}{\sqrt{\frac{n_1 n_2 (n_1 + n_2 + 1)}{12}}} \sim N(0,1)$$

Example: Two independent random samples of unemployed men and women are drawn and the ages of the 4 unemployed women and 5 unemployed men are recorded as follows:

Women:	60	63	36	44	
Men:	53	39	22	33	44

(i) Do these data present sufficient evidence to conclude that there is a difference in the average age of unemployed men and women?

(ii) Test whether the average age of unemployed women is significantly high.

Soln:

1. Formulation of hypothesis: $H_0: Md_1 = Md_2$

(a) $H_0: E(X) = E(Y)$ i.e., there is no difference between the average age of unemployed women and men.

(b) (i) $H_1: E(X) \neq E(Y)$ OR $H_1: Md_1 \neq Md_2$
(ii) $H_1: E(X) > E(Y)$ $H_1: Md_1 > Md_2$

2. Test statistic:
Calculation for test statistic U_0:

Combined ages in ascending order	Rank	Rank for 1st sample	Rank for 2nd sample
22	1	-	1
24	2	-	2
33	3	-	3
36	4	4	-
39	5	-	5
44	6	6	-
53	7	-	7
60	8	8	-
63	9	9	-
		$R_1 = 27$	$R_2 = 18$

As $n_1 = 4$ and $n_2 = 5$

$$U_1 = n_1 n_2 + \frac{n_1(n_1+1)}{2} - R_1 = 4\times5 + \frac{4(4+1)}{2} - 27$$
$$= 3$$

$$U_2 = n_1 n_2 + \frac{n_2(n_2+1)}{2} - R_2 = 4\times5 + \frac{5(5+1)}{2} - 18 = 17$$

$$\therefore U_0 = \min\{U_1, U_2\} = \min\{3, 17\} = 3$$

3. Critical value:
The probability associated with the value as small as the observed value $U_0 = 3$ for $n_1 = 4$ and $n_2 = 5$ is 0.0556
i.e., $P_0 = P(U \leq 3) = 0.0556$.

4. Decision:

(i) Since $2P_0 = 0.1112 > 0.05 = \alpha$
The data do not present sufficient evidence to reject the null hypothesis Ho. ~~accept Ho~~

(ii) Since $P_0 = 0.0556 > 0.05 = \alpha$, Accept Ho.

5. Conclusion:

(i) Hence, we conclude that there is no significant difference between the average age of unemployed women and men.

(ii) Hence we can conclude that the average age of unemployed women and men are equal.

Example: The amount of rainfall (in cm) of Nepal's two places Nagarkot and Phulchoki as revealed by the records of Meteorological division, during monsoon was as follows.

Nagarkot: 15 10 14 15 20 8 17 22 12 9

Phulchoki: 12 7 11 19 23 4 8 3 6 10
13 6 9 12 7 10 21 4 9 6
11 16 4

Do the data provide sufficient evidence to conclude that average amount of rainfall at Phulchoki is significant lower than that at Nagarkot?

Solution:

1. Formulation of hypothesis: Md_1, Md_2

(i) Ho: $E(X) = E(Y)$ i.e., the average amount of rainfall at Nagarkot and Phulchoki are equal where $E(X)$ and $E(Y)$ denote the average amount of rainfall at Nagarkot and Phulchoki respectively.

$\Rightarrow Md_1 < Md_2$

(b) H_1: $E(Y) < E(x)$ i.e., the average amount of rainfall at Phulchoki is significantly lower than that at Nagarkot (One tailed test).

Calculation for test statistic Z of Mann-Whitney U test

Combined amount of rainfall in ascending order	Rank	Rank for 1st sample	Rank for 2nd sample
3	1		1
4	2.5		2.5
4	2.5		2.5
6	5		5
6	5		5
6	5		5
7	7.5		7.5
7	7.5		7.5
8	9.5	9.5	
8	9.5		9.5
9	12	12	
9	12		12
9	12		12
10	15	15	
10	15		15
10	15		15
11	17.5		17.5
11	17.5		17.5
12	20		20
12	20	20	

12	20		20
13	22	~~22~~	22
14	23.5	23.5	
14	23.5		23.5
15	25.5	25.5	
15	25.5	25.5	
16	27		27
17	28	28	
19	29		29
20	30	30	
21	31		31
22	32	32	
23	33		33
		$R_1 = 221$	$R_2 = 340$

Now,

$$U_1 = n_1 n_2 + \frac{n_1(n_1+1)}{2} - R_1 = 10 \times 23 + \frac{10(10+1)}{2} - 221$$

$\Rightarrow U_1 = 64$

$$U_2 = n_1 n_2 + \frac{n_2(n_2+1)}{2} - R_1 = 10 \times 23 + \frac{23(23+1)}{2} - 340$$

$\Rightarrow U_2 = 166$

$\therefore U_0 = \min\{U_1, U_2\} = \{64, 166\} = 64$

Since $n_1 = 10$ and $n_2 = 23$, it is a large sample case and therefore the test statistic for Mann-Whitney U test is,

$$Z = \frac{U_0 - \mu_U}{\sigma_U} = \frac{U_0 - \frac{n_1 n_2}{2}}{\sqrt{\frac{n_1 n_2(n_1+n_2+1)}{12}}}$$

Since the ties are occurred between sample observation, the test should be corrected by applying a correction to the standard deviations σ_U and ______ the ties.

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So, corrected $\sigma_U = \sqrt{\left[\frac{n_1 n_2}{n(n-1)}\right]\left[\frac{n^3-n}{12} - T_i\right]}$

where,

$T_i = \frac{t_i^3 - t_i}{12}$ and t_i = number of times the ith rank is repeated.

Since the rank 9.5 is repeated 2 times, $t_1 = 2$

Similarly, $t_2 = 3$, $t_3 = 3$, $t_4 = 3$, $t_5 = 2$

$$\therefore \sum_{i=1}^{5} T_i = \sum_{i=1}^{5} \frac{t_i^3 - t_i}{12} = \frac{2^3-2}{12} + \frac{3^3-3}{12} + \frac{3^3-3}{12} + \frac{3^3-3}{12} + \frac{2^3-2}{12}$$

$$= 0.5 + 2 + 2 + 2 + 0.5 = 7$$

$$\therefore \text{corrected } \sigma_U = \sqrt{\left[\frac{n_1 n_2}{n(n-1)}\right]\left[\frac{n^3-n}{12} - \sum_{i=1}^{5} T_i\right]}$$

$$= \sqrt{\left[\frac{10 \times 23}{33(33-1)}\right]\left[\frac{33^3-33}{12} - 7\right]}$$

$$= 25.49$$

Therefore, the corrected test statistic is,

$$\text{corrected } Z = \frac{U_0 - \frac{n_1 n_2}{2}}{\text{corrected } \sigma_U} = \frac{64 - \frac{10 \times 23}{2}}{25.49} = -2$$

3. Critical Value:

$Z_\alpha = Z_{0.05} = -1.645$ (LTT)

$|Z_{0.05}|_{tab} = 1.645$

4. Decision:

$|Z|_{cal} = 2 > |Z|_{tab}$, Reject H_0, Accept H_1

5. Conclusion:

Hence the data provide sufficient evidence to conclude that average amount of rainfall at Phulchoki is significantly lower than that of Nagarkot at 5% level of significance.

Chi-Square Test

Application:

↳ Goodness of fit: The goodness of fit describes the differences between the observed and expected frequency distributions. The small differences between the observed and expected frequency distributions are assumed to be resulting from sampling error.

↳ Independent of attributes:

Example: Three samples are taken comprising of 120 university teachers, 150 doctors and 150 advocate. Each person chosen is asked to select one of the three categories that best represent his feelings towards a certain national policy. The categories are in favour of policy (F), against the policy (A) and different towards the policy (I). The results of the interview are given below:

Reaction

Sample:	F	A	I	Total
University Teachers	80	30	10	120
Doctors	70	40	40	150
Advocates	50	50	30	130

On the basis of these data, can it be concluded that the reactions or views of university teachers, doctors and advocates are homogeneous in so far a national policy under discussions is concerned?

Solution:

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1. Formulation of hypothesis:

(a) H_0: $F_X(x) = F_Y(x) = F_Z(x)$ for all x i.e., the three populations of university teachers, doctors and advocates are identical.

That is, three professional university teachers, doctors and advocates are not homogenous in giving their reactions on the national policy.

(b) H_1: $F_X(x) \neq F_Y(x) \neq F_Z(x)$ i.e., the three professional university teachers, doctors & advocates on the national policy.

2. Test Statistic:

Under H_0, the test statistic is,

$$\chi^2 = \sum_{i=1}^{3}\sum_{j=1}^{3} \frac{(O_{ij}-E_{ij})^2}{E_{ij}} \text{ is } \chi^2_{(3-1)(3-1)} = \chi^2_{(4)}$$

where $E_{ij} = \frac{R_i \times C_j}{n}$, $i=1,2,3$, $j=1,2,3$

Sample	Reactions F	A	I	Row Total R_i
University Teachers	80	30	10	120
Doctors	70	40	40	150
Advocates	50	50	30	130
Column Total (C_j)	200	120	80	$n=400$

$$E_{11} = \frac{R_1 \times C_1}{n} = \frac{120\times200}{400} = 60$$

$$E_{12} = \frac{R_1 \times C_2}{n} = \frac{120\times120}{400} = 36$$

$$E_{13} = \frac{R_1 \times C_3}{n} = \frac{120\times80}{400} = 24$$

$$E_{21} = \cancel{150} \frac{R_2 \times C_1}{n} = \frac{150\times200}{400} = 75$$

$E_{22} = \frac{150 \times 120}{400} = 45$, $E_{23} = \frac{150 \times 80}{400} = 30$

$E_{31} = \frac{130 \times 200}{400} = 65$, $E_{32} = \frac{130 \times 120}{400} = 39$

$E_{33} = \frac{130 \times 80}{400} = 26$

Therefore,

$$\chi^2 = \frac{(80-60)^2}{60} + \frac{(30-36)^2}{36} + \frac{(10-24)^2}{24} + \frac{(70-75)^2}{75}$$

$$+ \frac{(40-45)^2}{45} + \frac{(40-30)^2}{30} + \frac{(50-65)^2}{65} + \frac{(50-39)^2}{39} + \frac{(30-26)^2}{26}$$

$\chi^2 = 6.67 + 1 + 8.17 + 0.33 + 0.55 + 3.33 + 3.46 + 3.10 + 0.61$

$\chi^2_{cal} = 27.22$

3. Critical value:

The critical value of χ^2 at $\alpha = 5\%$ for $(3-1)(3-1)$ $= 2 \times 2 = 4$ degree of freedom is 9.46 i.e.,

$\chi^2_{0.05(4)} = 9.49$

4. Decision:

Since the computed χ^2 value, 27.22 is greater than the significant value 9.49, therefore, we reject H_0 and accept H_1.

5. Conclusion:

That means, all the professionals university teachers, doctors and advocates are not homogeneous in giving their reaction on the national policy.

Kruskal Wallis H - Test

Kruskal-Wallis H-test is a most powerful non-parametric alternative to the F-test for analysing data in one way classification. It is a non-parametric test of using ranks of observation in one way analysis of variation (one way ANOVA). This test is used to test whether the three or more populations differ in location (Md) significantly or to test whether the variation between $k(>2)$ independent random samples (treatments) is significant.

1. Formulation of hypothesis:

(a) H_0: $Md_1 = Md_2 = \dots = Md_k$ i.e., the k pop^n median Md_i $(i=1,2,\dots k)$ are equal.

(b) H_1: $Md_1 \neq Md_2 \neq \dots \neq Md_k$ i.e., the k population medians are not equal.

2. Test Statistic:

Under H_0,

$$H = \frac{12}{n(n+1)} \sum_{i=1}^{k} \frac{R_i^2}{n_i} - 3(n+1)$$

3. Critical Value:

For different values of n_i $(i=1,2,\dots,k)$ the probability P_0 associated with the value as extreme as observed, H is obtained from the Kruskal Wallish table. i.e., $P_0 = P(H \geq H_{cal})$

4. Decision:

If $P_0 \leq \alpha$ a pre-assigned level, we reject the null hypothesis H_0 and otherwise we accept the H_0 at the α-level.

If there are a substantial number of tied observations in a rank, an adjustment for Kruskal Wallis H test is to be made. The adjustment factor is,

$$C_f = 1 - \frac{\sum T_i}{n^3 - n}$$

where $T_i = t_i^3 - t_i$ being the number of tied observations in ith rank.

The corrected Kruskal Wallis H statistic is given by,

$$H_{corrected} = \frac{H}{C_f} = \frac{\frac{12}{n(n+1)} \sum \frac{R_i^2}{n_i} - 3(n+1)}{1 - \frac{\sum T_i}{n^3 - n}}$$

Case I:

Small sample case i.e., when $k < 3$ and $n_i \leq 5$.

8. Critical Value:

For different values of n_i ($i = 1, 2, 3 = k$), the probability P_0 associated with the value as extreme as observed H is obtained from the Kruskal-Wallis table i.e.,

$$P_0 = P(H \geq H_{cal})$$

9. Decision:

If $P_0 \leq \alpha$ pre-assigned level, reject H_0 and otherwise accept the H_0 at the α-level.

Case II:

Large sample case i.e., when $k \geq 3$ and $n_i > 5$.

In this case, the sampling distribution of H, under H_0, can be approximated by a chi-square distribution with $k-1$ degrees of freedom i.e H is $\chi^2_{(k-1)}$ under H_0.

3. Critical Value:
For a given α-level, we obtain the critical value of H from the chi-square table for $k-1$ degree of freedom i.e., $\chi^2_{\alpha}(k-1)$.

Example: The following are the numbers of misprints counted on pages selected at random from three editions of a book

Edition I	4	10	2	6	4	12
Edition II	8	5	13	8	8	10
Edition III	7	9	11	2	14	7

Use Kruskal Wallis H test at the 5% level of significance to test the null hypothesis that the samples came from identical populations.

Solution:

	No. of misprints						R_i	$\frac{R_i^2}{n}$
Edition I:	4	10	2	6	4	12		
Rank	3.5	13.5	1.5	6	3.5	16	= 44	= 322.6
Edition II:	8	5	13	8	8	10		
Rank	10	5	17	10	10	13.5	= 65.5	715.04
Edition III:	7	9	11	2	14	7		
Rank	7.5	12	15	1.5	18	7.5	= 61.5	630.37

$\sum \frac{R_i^2}{n_i} = \cancel{1667}$ 1668.076

Sample size of Edition I $(n_1) = 6$

Sample size of Edition II $(n_2) = 6$ $\quad n = n_1 + n_2 + n_3$

Sample size of Edition III $(n_3) = 6$ $\quad n = 6+6+6 = 18$

No. of times rank 1.5 is repeated $(t_1) = 2$

No. of times rank 3.5 is repeated $(t_2) = 2$

No. of times rank 7.5 is repeated $(t_3) = 2$

No. of times rank 10 is repeated $(t_4) = 3$

No. of times rank 13.5 is repeated $(t_5) = 2$

$$\therefore \Sigma(t_i^3 - t_i) = (2^3-2) + (2^3-2) + (2^3-2) + (3^3-3) + (2^3-2)$$
$$= 48$$

R_1, R_2, R_3 be sum of ranks for edition I, II and III respectively.

Let Md_1, Md_2 and Md_3 be median misprint of edition I, II and III respectively.

1. Formulation of hypothesis:

(a) H_0: $Md_1 = Md_2 = Md_3$ i.e., population are identical

(b) H_1: $Md_1 \neq Md_2 \neq Md_3$ i.e., population are not identical.

2. Test Statistic:

$$H = \frac{\frac{12}{n(n+1)} \Sigma \frac{R_i^2}{n_i} - 3(n+1)}{1 - \frac{\Sigma(t_i^3 - t_i)}{n^3 - n}}$$

$$H = \frac{\frac{12}{18(18+1)} \times 1668.076 - 3(18+1)}{1 - \frac{48}{18^3 - 18}}$$

$$H = \frac{58.528 - 57}{0.9917} = 1.5407$$

3. Critical Value:

Critical value at 5% level of significance for 2 degree of freedom is $\chi^2_{0.05(2)} = 5.99$

4. Decision.

$H = 1.54 < \chi^2_{0.05(2)} = 5.99$, Accept H_0, Reject H_1

5. Conclusion:

The samples come from identical population.

Friedman F- Test

Friedman F-test:

It is also Friedman two way ANOVA test. It is used to test the significance difference between location of three or more independent populations.

1. Formulation of hypothesis:

ⓐ H_0 : $Md_1 = Md_2 = Md_3 = \dots = Md_k$

ⓑ H_1 : At least one Md_i is different $i = 1, 2, 3, \dots, k$.

Rank k samples observation for each block separately from 1 to k in ascending order. If two or more observations are same then assign average rank which is also called tied. Obtain sum of ranks for each sample to get R_i. $i = 1, 2, 3, \dots, k$.

2. Test Statistic:

$$F_r = \frac{12}{nk(k+1)} \sum_{i=1}^{k} R_i^2 - 3n(k+1)$$

If tied occurs then corrected test statistic is,

$$F_r = \frac{\frac{12}{nk(k+1)} \sum_{i=1}^{k} R_i^2 - 3n(k+1)}{1 - \frac{\Sigma(t_i^3 - t_i)}{n(k^3 - k)}}$$

where t_i = number of times ith rank is repeated.

3. Critical Value:

For $\alpha = 5$, level of significance

For $2 \le n \le 9$ and $k = 3$, also, $2 \le n \le 4$ and $k = 4$. Critical value p is obtained from Friedman probability table.

For $n \geq 5$ and $k > 3$, critical value is $\chi^2_{\alpha}(k-1)$.

4. Decision:

Accept H_0 at α level of significance if $p > \alpha$, reject H_0 otherwise for $2 \leq n \leq 9$ and $k = 3$ also $2 \leq n \leq 4$ and $k = 4$.

Reject H_0 at α% level of significance if $H > \chi^2_{\alpha}(k-1)$, accept otherwise for $n \geq 5$ and $k > 3$.

Example: An investigator wants to study the scores of 3 matched groups under 5 conditions, each group contains five ~~gro~~ subjects, one being assigned to each of the five conditions. Let the score obtained are given in the following table. Is there any significant difference between three groups

Conditions

Group	I	II	III	IV	V
A	9	8	5	1	7
B	6	7	5	2	8
C	9	7	5	2	6

Solution:

Ranking the different conditions separately.

Conditions

Group	I	II	III	IV	V	R_i	R_i^2
A	2.5	3	2	1	2	10.5	110.25
B	1	1.5	2	2.5	3	10	100
C	2.5	1.5	2	2.5	1	9.5	90.25
							300.5

Here, Number of conditions $(n) = 5$

Number of groups $(k) = 3$

no. of times rank 2.5 is repeated in I $(t_1) = 2$

no. of times rank 1.5 is repeated in II $(t_2) = 2$

no. of times rank 2 is repeated in III $(t_3) = 3$

no. of times rank 2.5 is repeated in IV $(t_4) = 2$

Let Md_A, Md_B, Md_C be median of group A, B and C respectively.

1. Formulation of hypothesis:

(a) H_0: $Md_A = Md_B = Md_C$ i.e., scores of 3 matched group is same.

(b) H_1: $Md_A \neq Md_B \neq Md_C$ i.e., scores of 3 matched group is not same.

2. Test Statistic:

$$F_r = \frac{\frac{12}{n(k+1)} \sum R_i^2 - 3n(k+1)}{1 - \frac{\sum (t_i^3 - t_i)}{n(k^3 - k)}}$$

$$F_r = \frac{\frac{12 \times 300.5}{5(3+1)} - 3 \times 5 \times 4}{1 - \frac{(2^3-2)+(2^3-2)+(3^3-3)+(2^3-2)}{5(3^3-3)}}$$

$F_r = 0.153$

3. Critical Value:

Critical value is $P_0 = P(F_r > 0.153) = 0.4$ for $n=5$ and $k=3$

~~5. Condu~~

4. Decision:

$P_0 = 0.4 > \alpha = 0.05$, Accept H_0 at 5% level of significance.

5. Conclusion:

There is no significant difference between scores of three matched group.

Signed Pair Rank Test

Wilcoxon Matched Pair Signed Rank Test.

It is non-parametric test used to compare two populations for which observations are paired. It is based upon magnitude and direction of difference between observations within each pair of related random samples.

Let us consider two related samples of size n drawn from continuous populations with unknown medians Md_1 and Md_2 respectively. Let $x_1, x_2, \dots, x_n$ and $y_1, y_2, \dots, y_n$ be two related samples of size n.

1. Formulation of hypothesis:

(a) $H_0 : Md_1 = Md_2$ i.e. there is no significant difference between two population medians.

(b) $H_1 : Md_1 \neq Md_2$ (TTT)

OR $H_1 : Md_1 > Md_2$ (RTT)

$H_1 : Md_1 < Md_2$ (LTT)

Find $d_i = y_i - x_i$ or $x_i - y_i$ for each pair of observations (x_i, y_i), $i = 1, 2, \dots, n$.
R and d_i irrespective of sign in ascending order but omit $d_i = 0$. If two or more d_i are equal then assign the average rank and is called tied. In such case corrected sample $n_i = n - t$, t is number of tied occurred. Assign sign to the ranks with respect to the sign of d_i. Sum the ranks of + sign and − sign separately to get $s(+)$ and $s(-)$ respectively.

Finally we get,

$$T = \min\{S(+), S(-)\}$$

Case I: Small sample size ($n < 25$)

2. Test statistic:

$$T = \min\{S(+), S(-)\}$$

3. Critical value:

Obtain critical value from Wilcoxon Matched pair signed rank test table.

$T_{\alpha, n}$, where n is corrected sample size omitting $d_i = 0$.

4. Decision:

If $T_{cal} \leq T_{(\alpha, n)\, tab}$ Reject H_0 otherwise Accept H_0.

Case II:

For large sample size, the sampling distribution of T is approximately normally distributed with mean μ_T and variance σ_T^2.

$$\mu_T = \frac{n(n+1)}{4} \quad \text{and} \quad \sigma_T^2 = \frac{n(n+1)(2n+1)}{24}$$

2. Test statistic:

$$Z = \frac{T - \mu_T}{\sigma_T} = \frac{T - \frac{n(n+1)}{4}}{\sqrt{\frac{n(n+1)(2n+1)}{24}}} \sim N(0,1)$$

3. Critical value:

Obtain critical value of Z at $\alpha\%$ per assigned level of significance.

4. Decision:

If $|Z|_{cal} \le |Z|_{tab}$, Accept H_0

If $|Z|_{cal} > |Z|_{tab}$, Accept H_1

Example: The number of server manufactured from each of two companies A and B was recorded daily for a period of 10 days with the following results;

Days	1	2	3	4	5	6	7	8	9	10
A	172	165	206	184	174	142	190	169	161	200
B	201	179	159	192	177	170	182	179	169	210

Assuming both companies produced same daily output, test the hypothesis that there is no difference between median defectives for two companies against the alternative hypothesis that company B produced more defective than company A using Wilkcoxon Matched pairs signed rank test.

Sol^n:

Days	Company A(x_i)	Company B(y_i)	$d_i = y_i - x_i$	R_i
1	172	201	29	9
2	165	179	14	7
3	206	159	-47	-10
4	184	192	8	3
5	174	177	3	1
6	142	170	28	8
7	190	182	-8	-3
8	169	179	10	5.5
9	161	169	8	3
10	200	210	10	5.5

Here,

$n = 10$

sum of ranks of + sign $[S(+)] = 9+7+3+1+8+5.5$
$+3+5.5 = 42$

Sum of ranks of − sign $[S(-)] = 10+3 = 13$

Let Md_A and Md_B be median of company A and B respectively.

1. Formulation of hypothesis:

(a) $Md_A = Md_B$ i.e. there is no significant difference in median defect of company A and company B.

(b) $Md_A < Md_B$ i.e. there is significant in median defective in company B than company A.

2. Test Statistic:

$T = \min\{S(+), S(-)\} = \min\{42, 13\} = 13$

3. Critical Value: $T_{0.05}(10) = 8$

4. Decision: $T_{cal} = 13 > \underset{tab}{T_{0.05}}(10) = 11$

Accept H_0 at 5% level of significance.

5. Conclusion:

There is no difference between median defectives for two companies A and B.

Cochran Q-test

Cochran Q-test:

It is non-parametric test used for more than two related samples. It is used to test significant difference in frequencies or proportions of three or more related samples.

Let us consider K treatments $(K>2)$ applied to the set of n objects dichotomized as yes (Y) or no (N), present (P) or absence (A), pass (P) or fail (F), etc.

1. Formulation of Hypothesis:
   (a) H_0: All the treatments are equally effective
   (b) H_1: All the treatments are not equally effective

2. Test statistic:

$$Q = \frac{(K-1)\left[K\Sigma R_i^2 - (\Sigma R_i)^2\right]}{K\Sigma C_j - \Sigma C_j^2} \text{ or } \chi^2$$

where,

sum all the Y (positives) according treatments to get R_i (row wise) and according to objects to get C_j (column wise) $i = 1,2,3,\ldots K$ and $j = 1,2,3,\ldots n$. Then get, $\sum_{i=1}^{K} R_i$, $\sum_{i=1}^{K} R_i^2$, $\sum_{j=1}^{n} C_j$, $\sum_{j=1}^{n} C_j^2$

3. Critical Value:

$\chi^2_{\alpha(K-1)}$ is the critical value obtained according to level of significance and degree of freedom.

4. Decision:

If $Q_{cal} \leq \chi^2_{\alpha}(K-1)$ Accept H_0

If $Q_{cal} > \chi^2_{\alpha}(K-1)$ Reject H_1.

Example: Laptop users were asked for the acceptability of four brands for daily use. The response of acceptability (A) and rejection (R) are given below.

Users	Alfa	Beta	Gamma	Delta
H_1	A	R	A	R
H_2	R	A	A	R
H_3	R	A	R	A
H_4	A	R	R	R
H_5	A	A	R	A

Test whether there is any significant difference between brands with respect to acceptability.

Solⁿ:

Users

Brands	H_1	H_2	H_3	H_4	H_5	R_i	R_i^2
Alfa	A	R	R	A	A	3	9
Beta	R	A	A	R	A	3	9
Gamma	A	A	R	R	R	2	4
Delta	R	R	A	R	A	2	4
						$\Sigma R_i = 10$	$\Sigma R_i^2 = 26$
C_j	2	2	2	1	3	$\Sigma C_j = 10$	
C_j^2	4	4	4	1	9	$\Sigma C_j^2 = 22$	

Here,

Number of brands $(K) = 4$

Number of housewife $(n) = 5$

1. Formulation of hypothesis:

(a) H_0: There is no significant difference between brands.

(b) H_1: There is at least one significant difference between brands.

2. Test statistic:

$$Q = \frac{(k-1)\left[k\sum R_i^2 - \left(\sum R_i\right)^2\right]}{k\sum C_j - \sum C_j^2}$$

$$= \frac{(4-1)\left[4\times 26 - (10)^2\right]}{4\times 10 - 22}$$

$$= 0.666$$

3. Critical value:

$$\chi^2_{\cancel{tab}\ 0.05(3)} = 7.81$$

4. Decision:

Here, $Q < \chi^2_{0.05(3)}$, Accept H_0

5. Conclusion:

There is no significant difference between brands according to acceptability.

classmate

Date ______

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